

9) Week 15: Randomness

9.1) The variable X is uniformly distributed over the interval $[1, 3]$.

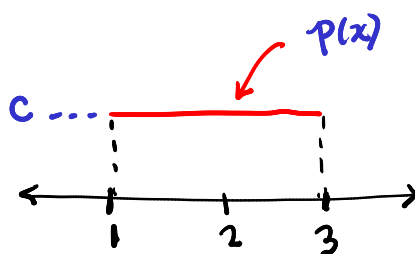
- Give a formula for the distribution of X .
- Compute the probability of the event $e = [\frac{3}{2}, \frac{5}{2}]$.

a)

• Random variable X has sample space $\Omega = [1, 3]$

• Uniform distribution:

$$c = \frac{1}{b-a} = \frac{1}{3-1} = \frac{1}{2}$$

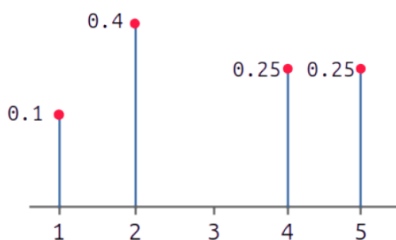


$$p(x) : [1, 3] \rightarrow \mathbb{R}$$
$$x \mapsto \frac{1}{2}$$

$$\begin{aligned} \text{b) } P\left(\left[\frac{3}{2}, \frac{5}{2}\right]\right) &= \int_{3/2}^{5/2} p(x) dx \\ &= \int_{3/2}^{5/2} \frac{1}{2} dx \\ &= \frac{1}{2} \end{aligned}$$



9.2) For the discrete distribution shown in the figure below, answer the following.



- What is its sample space?
- Calculate its mean.
- Calculate its standard deviation.
- What is the probability of the event $\{1, 2\}$?

$$a) \Omega = \{1, 2, 4, 5\}$$

$$b) \mu_x = \sum_{x \in \Omega} x p(x)$$

$$= 1(0.1) + 2(0.4) + 4(0.25) + 5(0.25) = 3.15$$

$$c) \sigma_x = \sqrt{E[(X - \mu_x)^2]}$$

Let's make a table for the r.v. $(X - \mu_x)^2$

x	$p(x)$	$(x - \mu_x)^2$
1	0.1	$(1 - 3.15)^2 = 4.62$
2	0.4	$(2 - 3.15)^2 = 1.32$
4	0.25	$(4 - 3.15)^2 = 0.72$
5	0.25	$(5 - 3.15)^2 = 3.42$

$$\sigma_x^2 = E[(X - \mu_x)^2] = 4.62(0.1) + 1.32(0.4) + 0.72(0.25) + 3.42(0.25) = 2.03$$

$$\therefore \sigma_x = 1.42$$

9.3) The function $p(x) = c x^2$ is defined on the interval $\Omega = [0, 3]$.

- a) What must the value of c be for $p(x)$ to be a valid distribution?
- b) Find the mean of X (the random variable whose distribution is $p(x)$).
- c) What is the probability of sampling a value of X in the interval $[1, 2]$?

a) Positivity: $p(x) \geq 0 \quad \forall x \in \Omega$

$$\Leftrightarrow c x^2 \geq 0 \quad \forall x \in [0, 3]$$

$$\Leftrightarrow c \geq 0$$

Unit integral $\int_{\Omega} p(x) = 1$

$$\Leftrightarrow \int_0^3 c x^2 = 1$$

$$\Leftrightarrow c \left. \frac{x^3}{3} \right|_0^3 = 1$$

$$\Leftrightarrow c \cdot 9 = 1 \quad \Leftrightarrow c = \frac{1}{9} \quad \checkmark$$

b) $\mu_x = \int_0^3 x p(x) dx$

$$= \int_0^3 \frac{1}{9} x^3 dx = \frac{1}{9} \left. \frac{x^4}{4} \right|_0^3 = \frac{81}{36} = \frac{9}{4}$$

c) $P([1, 2]) = \int_1^2 p(x) dx = \int_1^2 \frac{1}{9} x^2 dx = \frac{1}{9} \left. \frac{x^3}{3} \right|_1^2$

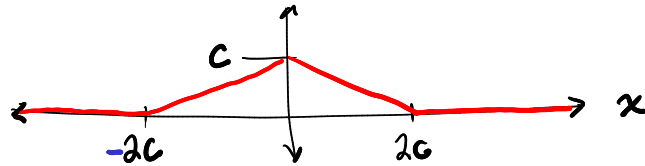
$$= \frac{8-1}{27} = \frac{7}{27}$$

9.4) The function $p(x)$ is defined on the real line.

$$p(x) = \max\left(0, c - \frac{|x|}{2}\right)$$

- Draw this function for $c = 1$.
- What must c be for $p(x)$ to be a valid distribution?
- With c set as in part b), what is the probability of the event $[-c, c]$?

a)



b) Positivity: $c \geq 0$

Unit integral: $\int_{-\infty}^{\infty} p(x) dx = 1$

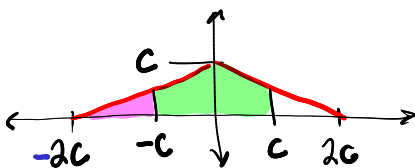
$$\Leftrightarrow \text{triangular area} = 1$$

$$\Leftrightarrow \frac{1}{2} \cdot 4c \cdot c = 1$$

$$\Leftrightarrow 2c^2 = 1 \Leftrightarrow c = \pm \sqrt{1/2} \quad \dots$$

Combining both
we have
 $c = \sqrt{1/2}$

c) $P([-c, c]) = \int_{-c}^c p(x) dx = \text{green area}$



$$= 1 - 2 \times (\text{pink area})$$

$$= 1 - 2 \left(\frac{1}{2} \cdot c \cdot \frac{c}{2} \right) = 1 - \frac{c^2}{2}$$

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

9.5) A Monte Carlo simulation has been run to estimate the variation of the temperature at the center of a heated plate when the temperatures at its corners are subjected to random variations. Upon running 3 executions of the code, we obtain the following dataset (expressed in degrees Celsius).

$$\mathcal{D} = \{51.7, 53.6, 52.9\}$$

- a) Compute the sample mean.
- b) Compute the sample standard deviation.
- c) Estimate the probability that the temperature falls between 51°C and 53°C .

$$n = 3$$

$$a) \hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{3} (51.7 + 53.6 + 52.9) = 52.7$$

$$b) \hat{\sigma}_n = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \hat{\mu}_n)^2}$$

$$= \sqrt{\frac{1}{2} \left((51.7 - 52.7)^2 + (53.6 - 52.7)^2 + (52.9 - 52.7)^2 \right)}$$

$$= 0.96$$

$$c) e = [51, 53]$$

$$\hat{p}(e) = \frac{\# \text{ samples in } e}{n} = \frac{2}{3}$$